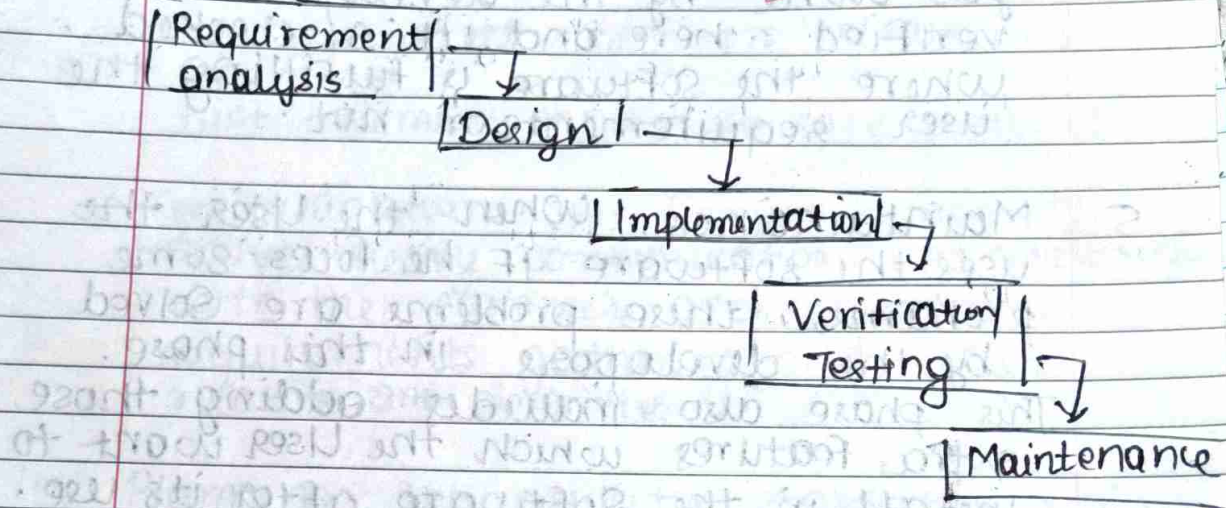


Waterfall model.



1. **Requirement Analysis** - This framework activity involves heavy communication and collaboration with customers.
2. **Design** : Based on these requirement Specification document design of the system is created and it is called Software architecture. It represents Blueprint of internal structure.
3. **Implementation** : This phase is actual coding which is constructed for Software architecture using hardware & Software requirements of System.

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4. Verification and Testing: Whatever job done by the developer is verified here and it is checked where the software is fulfilling the user requirements or not.

5. Maintenance: When the user uses the software if he faces some problems, these problems are solved by the developers in this phase. This phase also includes adding those extra features which the user wants to install in the software after its use.

Advantages:

Simple to Implement

Useful for Implementation of Small Systems.

Disadvantages

Real projects rarely follow this flow that the model processes

Iteration always occurs and creates problems in the application

Difficult for the Client to state all requirements explicitly.

Incremental Model.

In this model software releases in increments
first increment constitutes core product.

Communication: This framework activity involves heavy communication and collaboration with the customer and encompasses requirements gathering and other relative ~~and~~ activities.

Planning: This activity establishes a plan for the Software Engineering work that follows. It describes the technical tasks to be conducted, the risks that are likely, the resources that will be required, the work products to be produced and a work Schedule.

Modeling: This a activity encompasses the creation of models that allow the developer and the customer to better understand Software requirements and the design that will achieve those requirements.

Construction: This activity combines code generation (either manual or Automated) and the testing that is required to uncover errors in the code.

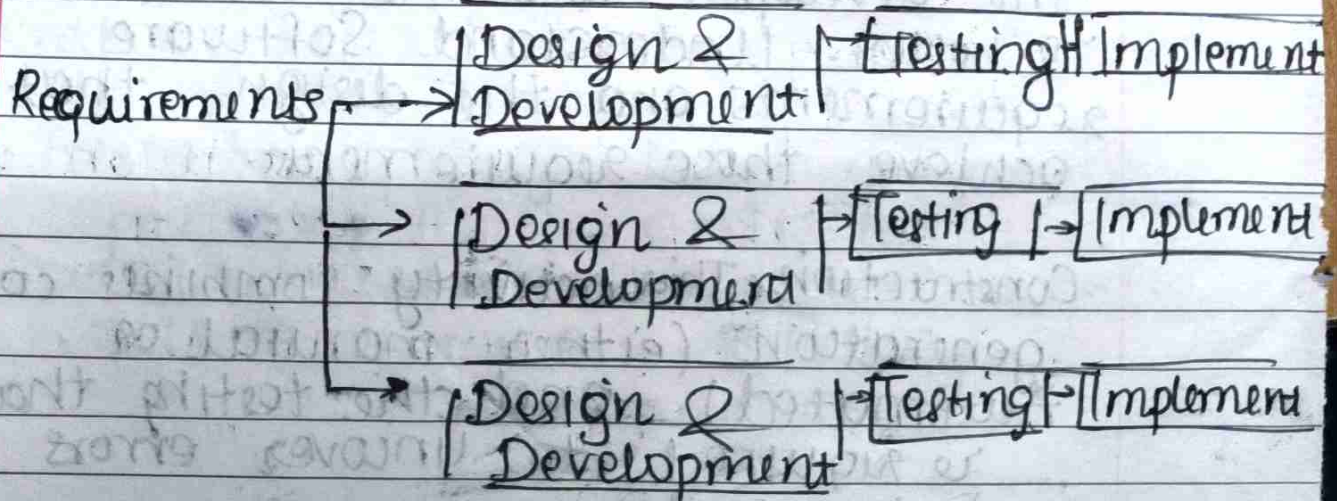
Deployment - The Software is delivered to the Customer who evaluates the delivered product and provides feedback based on the evaluation.

Advantages

- Particularly useful when staffing is inadequate for a complete implementation by the business deadline.
- Early installments or increments can be implemented with fewer people. If the core product is well received, additional staff can be added to implement the next increment.
- Increments can be planned to manage technical risks.

Disadvantages

As more functionality is added to the product, problems may arise or start related system architecture.



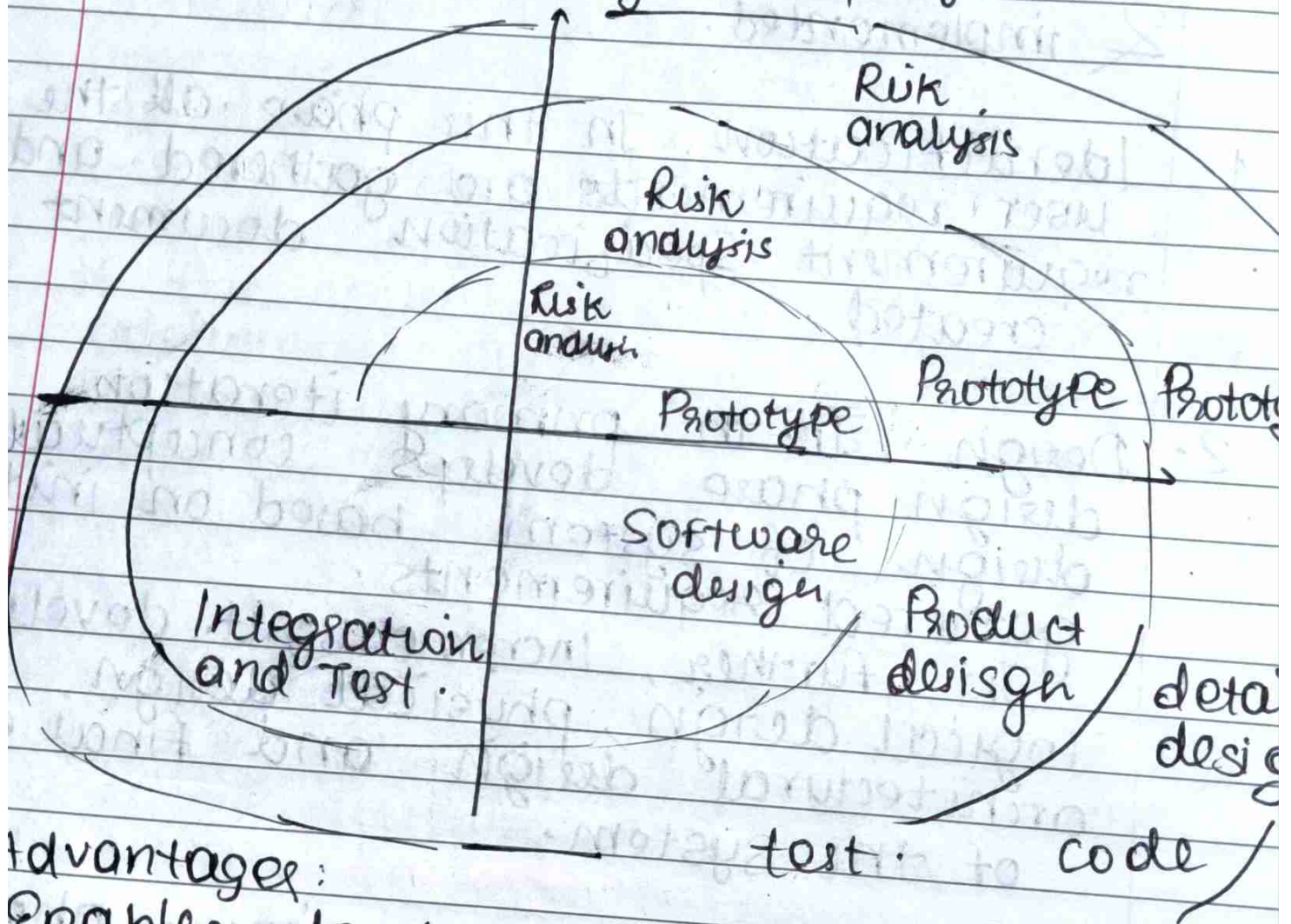
Spiral Model

Spiral model is a Risk driven process model. In Spiral model an alternate solution is provided if the risk is found in the risk analysis, then alternate solutions are suggested & implemented.

1. Identification: In this phase all the user requirements are gathered and a requirement Specification document is created.
2. Design: In the primary iteration design phase develops conceptual design of system based on initially gathered requirements.
In further increments, it develops logical design, physical design, architectural design and final design of the system.
3. Construction: In the primary phases code for conceptual design is developed and gets feedback from client. In next subsequent spirals, detailed working model of software is constructed with incremental numbers and are decided to customer for feedback.

7. evaluation and risk Analysis
In this phase management of risks like Cost over run are noticed and handled.

8.5. Deployment: After the final step the software gets deployed.



Advantages:

Enables developers and clients to understand and react to risks at an evolutionary level.

It uses the systematic approach suggested by the classical lifecycle method but incorporates it into an iterative framework that is more realistic.

This model demands an evaluation of risks at all stages and should reduce risks before they become problematic if properly applied.

Disadvantages:

1. It may be difficult to convince clients that the evolutionary approach is controllable.
2. It demands considerable risks assessment expertise and relies on this for success.

Agile Methodology.

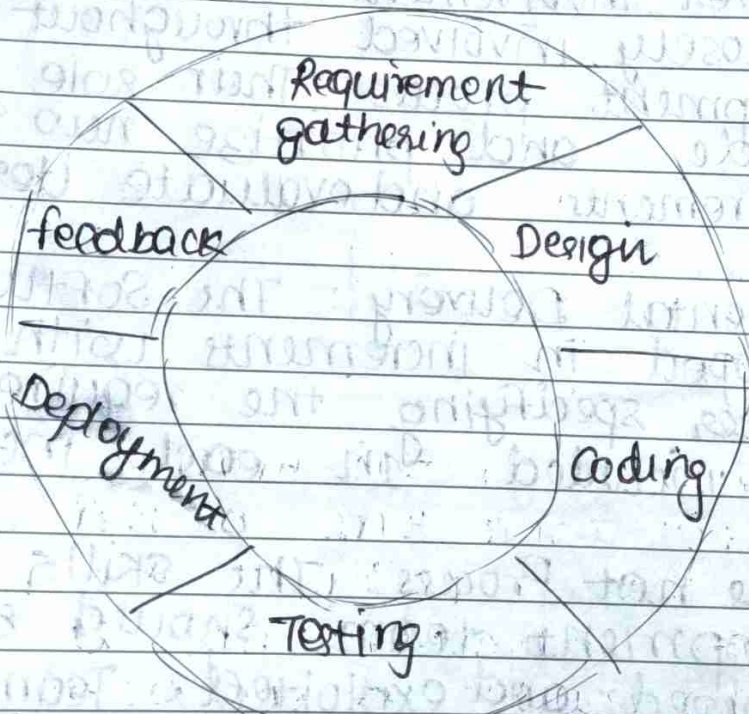
1. Customer Involvement: Customer should be closely involved throughout the development process. Their role is provide and prioritize new system requirements and evaluate iterations.
2. Incremental Delivery: The software is developed in increments with the customer specifying the requirements to be included in each increment.
3. People not Process: The skills of the development team should be recognized and exploited. Team members should be left to develop their own ways of working.

4. Embrace Change: Expect the System requirements to change and so design the system to accommodate these changes.

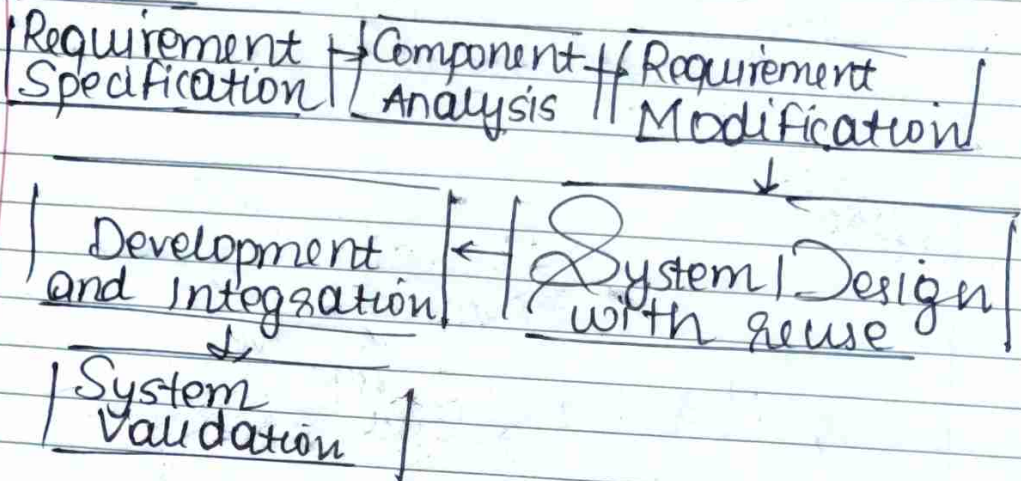
5. Maintain Simplicity: focus on simplicity in both the software being developed and in the development process.

Advantage.

- Customer involvement is continuous
- faster software is delivered early.
- Easy to adapt.
- continuous testing and integration



Component Based Models



The Component Based Software Models built Software by assembling pre-built reusable component instead of writing

Advantages

- Some Components can be used in Multiple Projects
- Reduced Development Time
- Improved Reliability
- Easy Maintenance

Disadvantages

- components may not fit perfectly
- Pre-built components may not fit perfectly together
- Limited Customization
- Performance Overhead
- Compatibility Problems